

ex. 1

5.3 MeV proton moves ^{+y} upward enters $B = 0.0012 \text{ T}$ directed ^{+z} outward. The magnetic force is:

$$KE = 5.3 \times 10^6 \times 1.6 \times 10^{-19} = 8.48 \times 10^{-13} \text{ J}$$

$$\begin{array}{l|l} K = 10^3 & m = 10^{-3} \\ M = 10^6 & \mu = 10^{-6} \\ G = 10^9 & n = 10^{-9} \end{array}$$

$$v = \sqrt{\frac{2KE}{m}} = \sqrt{\frac{2 \times 8.48 \times 10^{-13}}{1.67 \times 10^{-27}}} = 31.9 \times 10^6 \text{ m/s}$$

$$F_B = qvB = 1.6 \times 10^{-19} \times 31.9 \times 10^6 \times 0.0012 = 6.125 \times 10^{-15} \text{ N } (+x)$$

ex. 2

A proton moves with $\vec{v} = 6\hat{j} + 2\hat{k}$ enters $\vec{B} = 3\hat{j} + 5\hat{k}$. The magnetic force is:

$$\vec{v} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 6 & 2 \\ 0 & 3 & 5 \end{vmatrix} = \hat{i}(30-6) - \hat{j}(0-0) + \hat{k}(0-0) = 24\hat{i}$$

$$\vec{F}_B = q \vec{v} \times \vec{B} = 1.6 \times 10^{-19} \times 24 = 38.4 \times 10^{-19} \text{ N}$$

مسألة الوقت (دقيقتين)

ex. 3

0.075 m wire lies at +x, $I = 2.4$ A
in $B = 1.6$ T at +z. The magnetic
force F_B is:

$$F_B = I \times B = 2.4 \times 0.075 \times 1.6 = 0.288$$

$$F_B = 0.288 \text{ N at } -y$$

ex. 5

A proton enters $B = 0.42$ T

The time to complete one cycle is:

$$T = \frac{2\pi m}{qB} = \frac{2\pi \times 1.67 \times 10^{-27}}{1.6 \times 10^{-19} \times 0.42}$$

$$T = 1.56 \times 10^{-7} \text{ sec.}$$

Q8

A Proton moves with 2×10^6 m/s
enters $B = 0.42$ T at $\theta = 30^\circ$ at +x
The distance travelled in one cycle

$$T = \frac{2\pi m}{qB} = \frac{2\pi \times 1.67 \times 10^{-27}}{1.6 \times 10^{-19} \times 0.42} = 1.56 \times 10^{-7} \text{ sec.}$$

$$x = 2 \times 10^6 \times \cos 30^\circ \times 1.56 \times 10^{-7} = 0.27 \text{ m}$$

ex. 6

An electron accelerated from rest by 350 V ^{Vp}

The Speed is:

$$v = \sqrt{\frac{2qV_p}{m}} = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 350}{9.11 \times 10^{-31}}} =$$

Q11

An Proton accelerated from rest
by 120 V. The Speed is:

$$v = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 120}{1.67 \times 10^{-27}}} = 0.15 \times 10^6 \text{ m/s}$$